

Yang Qu

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EDUCATION

2013 – 2017 Ph.D. in Animal Science, **University of Maryland**, College Park, MD
2022 – 2024 M.Sc. in Mathematics, **University of North Carolina at Charlotte**, Charlotte, NC
2011 – 2013 M.Sc. in Animal Science, **Oregon State University**, Corvallis, OR
2006 – 2011 B.Sc. in Veterinary Science, **China Agricultural University**, Beijing, China

RESEARCH EXPERIENCE

Research Associate (Biostatistics) (Full-time) May 2024 – Present
College of Veterinary Medicine, North Carolina State University, Raleigh, NC

- Provide comprehensive statistical consulting and research support for veterinary and biomedical studies conducted at CVM at NCSU, including study design (randomized controlled trial, retrospective/prospective study, cross-over design, etc.), statistical analysis (e.g., cross-sectional data, Longitudinal data), and interpretation of results for both clinical investigations and laboratory-based research.
- Provide support on preparing statistical analysis methods for manuscript publication and statistical analysis plans (e.g., power analysis, sample size estimation, etc.) for research grant applications for internal and external funding opportunities.

Graduate Research Assistant (Part-time) 2022 – 2023
Department of Mathematics and Statistics, University of North Carolina at Charlotte, Charlotte, NC

- Led a project of Bayesian computation in small area estimation using R.
- Prepared descriptive statistics analysis on public health data.

Summer Research Internship (Part-time) Summer 2023
College of Computing and Informatics, University of North Carolina at Charlotte, Charlotte, NC

- Performed dimension-reduction analyses, including principal component analysis, and regression modeling on a high-dimensional microbiome dataset using R.
- Performed literature review on comparative genomic analysis, focusing on the theory of methods and its applications.

Postdoctoral Fellow (Full-time) 2017 – 2019
Department of Animal and Avian Sciences, University of Maryland, College Park, MD

- Designed animal experiments with a full factorial structure.
- Conducted molecular analysis, including qPCR, RNA, and microbiome sequencing.
- Completed data analysis using R and published results in the Journal of Dairy Science.

Graduate Research Assistant (Full-time) 2013 – 2017
Department of Animal and Avian Sciences, University of Maryland, College Park, MD

- Conducted *in vitro* experiments on bovine neutrophil metabolism using Gas Chromatography-Mass Spectrometry.
- Assessed the impact of short-term dietary tocopherol supplementation on tissue absorption, immune function, and gene expression in lactating dairy cows.

- Analyzed gene expression related to tocopherol metabolism in bovine liver tissue cells and the mitochondria.
- Investigated the effects of tocopherol supplementation on inflammatory responses and immune function in isolated bovine neutrophils.

TEACHING EXPERIENCE

Invited Lecturer (Veterinary Scholars Program)

June 2025

College of Veterinary Medicine, North Carolina State University, Raleigh, NC

- Held a one-hour invited lecture titled “*Statistics for Beginners*” to first-year veterinary medicine students participating in the 2025 Veterinary Scholars Program.
- Taught foundational concepts in statistical thinking, study design, and data interpretation using real veterinary research examples.

Graduate Teaching Assistant (College Algebra Workshops)

2023 – 2024

Department of Mathematics and Statistics, University of North Carolina at Charlotte, Charlotte, NC

- Conducted four 75-minute College Algebra workshop sessions independently each week, with approximately 25 registered undergraduate students per session.
- Held office hours twice per week to provide individualized student support.
- Participated in weekly College Algebra instructional team meetings.

Mathematics and Statistics Tutor

2022 – 2023

Department of Mathematics and Statistics, University of North Carolina at Charlotte, Charlotte, NC

- Provided tutor service for undergraduate students in College Algebra, Calculus I, Calculus II, and Elementary Statistics.
- Held regular office hours independently per week.

Graduate Teaching Assistant (Animal Nutrition)

2013 – 2017

Department of Animal and Avian Sciences, University of Maryland, College Park, MD

- Held lectures on animal nutrition to undergraduate students two times each week.
- Taught two animal nutrition laboratory sessions per week (approximately 20 students per session, 60 minutes each).
- Held office hours and graded students’ homework and exams.

PUBLICATIONS

1. Qu, Y., Lytle, K., Traber, M.G., and Bobe, G. (2013). Depleted serum vitamin E concentrations precede left displaced abomasum in early-lactation dairy cows. *Journal of Dairy Science*, 96(5), 3012–3022. doi:10.3168/jds.2012-6252
2. Qu, Y., Fadden, A.N., Traber, M.G., and Bobe, G. (2014). Potential risk indicators of retained placenta and other diseases in multiparous cows. *Journal of Dairy Science*, 97(7), 4151–4165. doi:10.3168/jds.2013-7814
3. Qu, Y., Elsasser, T.H., Kahl, S., Garcia, M., Scholte, C.M., Connor, E.E., Schroeder, G.F., and Moyes, K.M. (2018). The effects of feeding mixed tocopherol oil on whole blood respiratory burst and neutrophil immunometabolism-related gene expression in lactating dairy cows. *Journal of Dairy Science*, 101(5), 4332–4342. doi:10.3168/jds.2017-13555

4. Qu, Y., Kahl, S., Miska, K.B., Schreier, L.L., Russell, B., Elsasser, T.H., and Proszkowiec-Weglarz, M. (2021). The effect of delayed feeding post-hatch on caeca development in broiler chickens. *British Poultry Science*, 62(5), 731–748. doi:10.1080/00071668.2021.1925554
5. Oeffner, S.P., Qu, Y., Just, J., Quezada, N., Ramsing, E., Keller, M., Cherian, G., Goddick, L., and Bobe, G. (2013). Effect of flaxseed supplementation rate and processing on the production, fatty acid profile, and texture of milk, butter, and cheese. *Journal of Dairy Science*, 96(2), 1177–1188. doi:10.3168/jds.2012-6084
6. Garcia, M., Elsasser, T.H., Juengst, L., Qu, Y., Bequette, B.J., and Moyes, K.M. (2016). Amino acid supplementation and stage of lactation alter apparent utilization of nutrients by blood neutrophils from lactating dairy cows in vitro. *Journal of Dairy Science*, 99(5), 3777–3783. doi:10.3168/jds.2015-10778
7. Garcia, M., Elsasser, T.H., Qu, Y., Zhu, X., and Moyes, K.M. (2015). Glucose supplementation has minimal effects on blood neutrophil function and gene expression in vitro. *Journal of Dairy Science*, 98(9), 6139–6150. doi:10.3168/jds.2014-9251
8. Garcia, M., Qu, Y., Scholte, C.M., O'Connor, D., and Rounds, W.P. (2017). Regulatory effect of dietary intake of chromium propionate on the response of monocyte-derived macrophages from Holstein cows in mid lactation. *Journal of Dairy Science*, 100(8), 6389–6399. doi:10.3168/jds.2016-11966
9. Qu, Y. (2017). *The effect of feeding mixed tocopherol oil on body accumulation and immune cell functions in lactating Holstein dairy cows*. Doctoral dissertation, University of Maryland, College Park.
10. Scholte, C.M., Nelson, D.C., Garcia, M., Linden, S.B., Elsasser, T.H., Kahl, S., Qu, Y., and Moyes, K.M. (2018). Recombinant bacteriophage endolysin PlyC is nontoxic and does not alter blood neutrophil oxidative response in lactating dairy cows. *Journal of Dairy Science*, 101(7), 6419–6423. doi:10.3168/jds.2017-13997
11. Qu, Y., Elsasser, T.H., Newbold, J.R., Connor, E.E., Garcia, M., Scholte, C.M., and Moyes, K.M. (2016). Short-term feeding of a tocopherol mix (α -, β -, γ -, and δ -) alters the daily pattern of tocopherol isoforms present in milk and blood in lactating dairy cows. *Journal of Animal Science*, 94(Suppl. 5), 810–810. doi:10.2527/jam2016-1631

SKILLS

Statistical Software: R, SAS, JMP, GraphPad

PROFESSIONAL SERVICE

- Invited Reviewer, *Journal of Dairy Science* (2019)

ACTIVE MEMBERSHIP

- Member, American Statistical Association (ASA)
- Member, National Institute of Antimicrobial Resistance Research and Education (NIAMRRE)

HONORS AND AWARDS

- Graduate Dean's Fellowship, University of Maryland, College Park (2013)
- 1st Place Poster Award, Mid-Atlantic Nutrition Conference, Timonium, MD (2013)

- Jacob K. Goldhaber Travel Award, University of Maryland, College Park (2013)
- TLTC International Teaching Fellowship, University of Maryland, College Park (2015)
- Graduate Student Summer Research Fellowship, University of Maryland, College Park (2014)
- Graduate Teaching Assistantship, University of North Carolina at Charlotte (2023)